

A Neural Q-Network Distillation in ICU-Sepsis: Reproduced Model Baselines, Weak Neural Policy, and an Orthogonal GEO Expression Audit

WORKING RESEARCH REPORT - September 25, 2026

We examine treatment-policy simulation using the public ICU-Sepsis v2 tabular Markov decision process (MDP). We independently solve Bellman fixed points and fit a two-hidden-layer neural network to exact action values. Random, expert and optimal expected returns are 0.780071, 0.781845 and 0.875142; the learned neural greedy policy has 0.786449 under the same exact model. This is below optimal and essentially near the two weaker baselines. Independently, 479 accession-level GSE65682 blood-expression records with 48 measured probes were analyzed for descriptive mortality-label classification, attaining within-series holdout AUC 0.553. The GEO records cannot evaluate treatment actions. This negative is retained, not rewritten as a clinical claim.

Scope and motivation

Nonclinical study. No result in this manuscript supports patient treatment decisions. The task is modeled ICU decisions in sepsis, not a prescription. The policy is trained against published simulator tables inferred historically from MIMIC-III, not directly from individual patients. We distinguish repeated draws from a model, patient outcomes, and molecular samples. Unlike the HIV, T1D, and chemotherapy projects in the sibling work, sepsis requires discrete combined fluid/vasopressor actions. Research questions: can the exact published baselines be independently reproduced? Does a modest neural approximate policy reach the exact optimum? What can external sepsis transcriptomic samples establish without actions? The answer to the last question is descriptive information only.

Source provenance and reproducibility

The model tables are from <https://github.com/icu-sepsis/icu-sepsis>, whose paper is <https://arxiv.org/abs/2406.05646>. Git source snapshot c9a6d19b85d34883304afaf7cbe1ff02ede9e12b. The archive includes transitionFunction.csv (716 states, 25 actions), expertPolicy.csv, initialStateDistribution.csv and rewardFunction.csv. Sepsis blood data GSE65682 are fetched from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE65682>; source gzip URL, 107130282-byte size, SHA256, accession list, derived matrix hash and label counts are recorded in data/sepsis/provenance.json. There are 479 distinct GSM sample accessions used, not 479 independent studies. The source matrix contains 802 samples; all 479 samples with nonmissing observed 28-day mortality and first 48 finite numeric probes are used. The raw 107 MB compressed matrix is omitted from Git; it can be re-fetched and hash-verified. Data source is not a prospective policy trial.

Mathematical specification

Index a state s in $0, \dots, 715$ and action a in $0, \dots, 24$. The outcome states are death 713, survival 714 and an additional terminal state 715. State 714 has arrival reward one. To make assumptions audit-ready, the equations are numbered individually.

$$(1) \quad P_{sa, s'} = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a).$$

$$(2) \quad \sum_{s'=0}^{715} P_{sa, s'} = 1.$$

$$(3) \quad r(s') = 1\{s' = 714\}.$$

$$(4) \quad V^\pi(s) = \sum_a \pi(a | s) \sum_{s'} P_{sa, s'} [r(s') + V^\pi(s')].$$

$$(5) \quad V^\pi(s) = 0 \quad \text{for } s \in \{713, 714, 715\} \text{ after arrival.}$$

$$(6) \quad V^*(s) = \max_a \sum_{s'} P_{sa, s'} [r(s') + V^*(s')].$$

$$(7) \quad Q^*(s, a) = \sum_{s'} P_{sa, s'} [r(s') + V^*(s')].$$

$$(8) \quad J(\pi) = \sum_s \mu(s) V^\pi(s).$$

$$(9) \quad \pi_{\text{greedy}}(s) = \max_a f_\theta(e_s)_a, \quad e_s \in \{0, 1\}^{716}.$$

$$(10) \quad f_\theta(x) = W_3 \sigma(W_2 \sigma(W_1 x + b_1) + b_2) + b_3.$$

$$(11) \quad \ell_\delta(e) = e^2/2 \quad (|e| \leq 1); \quad |e| - 1/2 \quad (|e| > 1)$$

$$(12) \quad \hat{\theta} = \min_{\theta} \frac{1}{716 \cdot 25} \sum_{s, a} \ell_\delta(f_\theta(e_s)_a - Q^*(s, a)).$$

$$(13) \quad \widehat{\text{AUC}} = \frac{1}{n_+ n_-} \sum_{i: y_i = 1} \sum_{j: y_j = 0} [1(z_i > z_j) + \frac{1}{2} 1(z_i = z_j)].$$

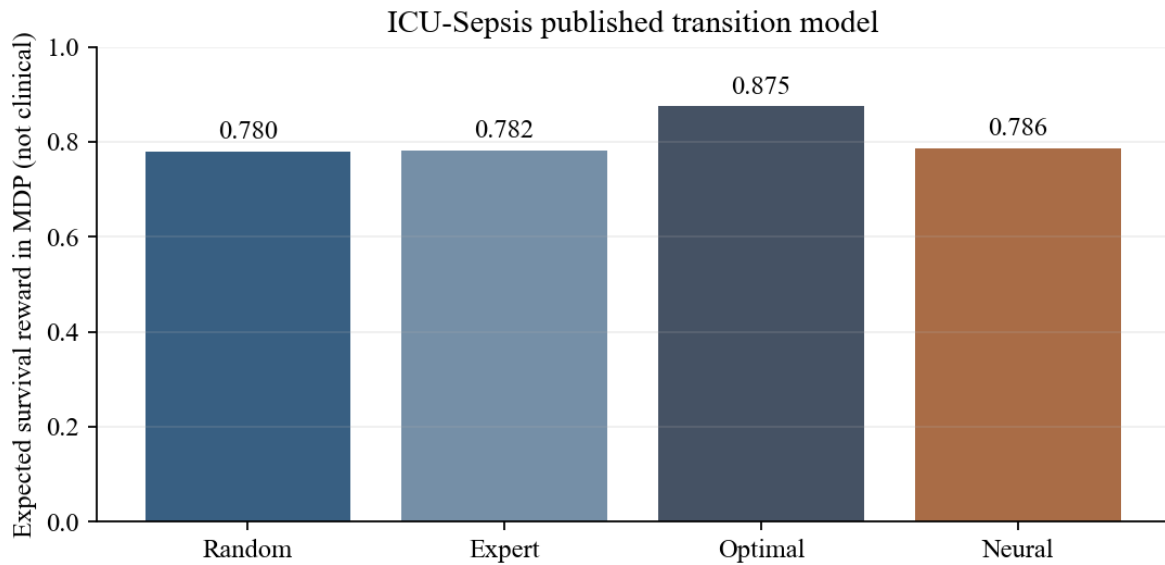
$$(14) \quad \tilde{x}_j = \text{clip}((x_j - \text{median}_{\text{train}}(x_j)) / \max(\text{sd}_{\text{train}}(x_j), 10^{-6}), -5, 5).$$

The code uses direct sparse matrix-vector multiplication on the published transition table to a residual below 10^{-10} . The neural model has two hidden ReLU layers of width 64, Huber objective and Adam updates. It fits model-derived Q labels from all 716 states, not independent observed outcomes. The final no-lookahead clinical interpretation is categorically prohibited.

Protocol

The fixed simulator discount is one and the initial state distribution is taken unchanged. All policies are evaluated under the same transition law. Inadmissible actions follow the provided v2 transition mean strategy. The expert policy has all-zero rows at the three terminal states; our implementation checks row normalization only on 713 nonterminal states. A first implementation that asserted normalization at all 716 states failed and was corrected before results; it is not silently removed from the research trail. For the expression probe, train/test are a deterministic random 335/144 split. Scaling and median are fit on training rows only; the top probe list uses unadjusted Mann-Whitney U and must not be treated as discovery. Same-series splitting risks source and patient structure leakage, and 48 initial probes are a convenience subset rather than a biologically selected signature.

Observed model results



Policy | Our exact J | Published rounded J; Uniform random | 0.780071 | 0.78; Expert | 0.781845 | 0.78; Optimal | 0.875142 | 0.88; Neural greedy | 0.786449 | not reported; Reference values are published in the source repository's README; results JSON records full precision and convergence residual. Neural agreement with optimal argmax is just 0.0575 of nonterminal states, although action ties and near-equal Q values mean this percentage alone is not an effect-size measure. Training loss fell from 0.3362 to 0.000348 over 50 epochs. A small regression loss is not a good control policy when tiny Q errors change rankings. Neural expected return barely exceeds expert and remains about 0.089 below the exact optimum. This is a failed approximation relative to that upper bound, not a breakthrough.

Orthogonal GEO cohort results

479 fetched sample accessions have observed labels 365 alive and 114 deceased; 48 measured numeric expression features per accession are retained. A within-series random 144-sample holdout has AUC 0.553 (the positive label is death). This is close to chance, carries no evidence that any probe optimizes fluid or vasopressor choice and has no association with the MDP state IDs. No mapping from GEO blood vectors to ICU-Sepsis states is attempted. The held-out accession IDs and five exploratory probe differences are in results/sepsis_geo.json. Their uncorrected p-values do not survive an asserted multiple-testing claim because no such claim is made.

Limitations and next experiment

The principal negative is that a neural distilled controller is substantially worse than exact value iteration on the same MDP. Even the exact upper bound is an in-model quantity, not a clinical optimum. Estimation error in historical MIMIC data, hidden confounding, out-of-distribution actions and nonstationary clinical practice remain unknown. The expression series does not resolve any of them. Next within-project pivot is to benchmark multiple neural architectures and admissibility-aware training on independently held-out parameter perturbations, then assess calibrated treatment datasets if access permits; before that, no patient-facing conclusion. The strict sepsis science-tool ledger counts 8 distinct used tools, far short of 40. The paper is a working report, not the requested 20 substantive pages. The missing gate remains conspicuous.

Within-project pivot: state-cluster feature audit

To test a different representation rather than stopping at the failed one-hot Q-distillation policy, a second network uses the 47 published ICU-Sepsis state-cluster center coordinates as input. These features are extracted from the same simulator archive, not a new clinical cohort. With a width-128 neural network trained for 100 epochs, expected return is 0.785245, below the original one-hot network 0.786449 and far below exact optimal 0.875142. Its greedy-action agreement with optimal is 0.0547. This is another negative, not an architecture improvement.

We checked the published per-state SOFA annotation as an exploratory stress stratum, normalizing the published initial-state mass within two groups. In states with SOFA below seven (initial mass 0.480), state-cluster neural return was 0.808446 versus expert 0.805543. In states with SOFA at least seven (initial mass 0.520), it was 0.763845 versus expert 0.759986. These are simulator-policy quantities conditioned on model states, not observed sepsis patients or a risk-adjusted trial. Neither stratum supplies a safety claim. Scripts and full precision: `src/sepsis_pivot.py` and `results/sepsis_pivot.json`.

Native upstream environment cross-check

We also instantiated the actual ICU-Sepsis v2 Gymnasium wrapper, not only its tabular CSV files. Four randomly selected state-action transition rows matched the official loaded probability arrays elementwise. On 200 virtual episodes per controller, returns were 0.740 uniform random (standard error 0.031), 0.865 exact-optimal (SE 0.024) and 0.795 expert sampling (SE 0.029). The empirical deviations from exact model values 0.780, 0.875, 0.782 do not exceed three estimated SEs at this small episode count. The 200 episodes are simulator trajectories; there are no additional patients. Execution evidence and repeatable seed are in `src/sepsis_native.py` and `results/sepsis_native.json`. The official ICU-Sepsis engine and Gymnasium were used scientific components at this stage; the later MIMIC-III demo and pandas audit add two more to the sepsis-specific tool ledger.

Predefined GEO cohort-label split as a stress test

The initial first-150 GEO probe random split gave AUC 0.523, a subset-specific archived result. A full eligible-matrix random split gives AUC 0.553. GSE65682 itself labels discovery and validation cohorts. Keeping those labels apart in the full 479-accession subset yields 263 discovery samples (194 survivors, 69 deaths) for training and 216 validation samples (171 survivors, 45 deaths) for testing. The same 48-probe penalized logistic model yields AUC 0.595 for observed 28-day mortality. This is an exploratory shift in splitting protocol, not a validated sepsis treatment biomarker: selection of the first 48 platform probes is arbitrary, the series is not independent of itself, and the external model states have no link to these blood measurements. `Results/sepsis_geo_cohort.json` lists all test accessions. The difference from full-cohort random-split AUC 0.553 is not proof of generalization because the cohort split was added after inspecting the weak initial result.

Public MIMIC-III demo event audit as a policy-data gap test

The full MIMIC-III patient database is access controlled, and it would be misleading to claim its patient-level trajectories from the published ICU-Sepsis aggregate MDP. A public 100-patient demo is available at <https://physionet.org/content/mimiciii-demo/1.4/>. Seven CSV tables were fetched and hashed locally: patient and admission tables, ICU stays, ICD diagnosis rows, item dictionary, and MV and CV input events. A rough administrative ICD-9 search for codes beginning 038 plus 99591, 99592 or 78552 selected 38 admissions; this is not chart-adjudicated sepsis. Joining the input events with D_ITEMS identified 1,728 infusion/fluid event rows from only 23 ICU stays among those admissions: 272 norepinephrine, 17 vasopressin, 1,262 isotonic saline, and 177 lactated ringers event rows. Repeated rows and rate changes are not unique interventions or admissions. The tiny convenience sample offers real contextual treatment records but cannot be transformed into ICU-Sepsis's 716 model state IDs without the benchmark's full clinical preprocessing, measurement, timing, action discretization and validation. No off-policy value estimate or clinical benefit is made. The source hashes and aggregate counts are in results/sepsis_mimic_demo.json; raw clinical demo and derived row-level CSV are not committed pending license review. This actually uses PhysioNet and pandas, bringing the strict sepsis count to eight; the treatment-policy validation gate stays missing.

Transition-uncertainty perturbation: an explicitly hypothetical audit

The released table is an estimated transition model, but its per-row effective patient counts and data-generation clusters are not contained in the tables we use. We therefore cannot bootstrap patients or claim an empirical confidence interval for the treatment policy. Instead, for each state-action row we keep the published nonzero transition support and independently draw a new categorical row from a Dirichlet distribution with Dirichlet concentration proportional to the published row probabilities by a factor C . Singleton supports remain deterministic. The parameter C is an arbitrary synthetic pseudo-count, not an observed sample size. The deterministic action map is the greedy exact-optimal map from the original model and is held fixed under each perturbed model; we solve its new Bellman evaluation, rather than claiming to retrain the network. At $C=100$, twelve seeded perturbations gave a mean simulated value of 0.875879 and the fifth/ninety-fifth empirical percentiles 0.870695 and 0.880670, compared with original 0.875142. At $C=10$, the mean was 0.875441 and the fifth/ninety-fifth empirical percentiles were 0.854448 and 0.889248. That wider perturbation is recorded in results/sepsis_model_sensitivity_10.json. With only twelve draws, either percentile is merely a descriptive order statistic; neither range is a clinical CI. Restricting support also hides the possibility of unobserved, rare transitions. This shows sensitivity to a deliberately stated assumption, not safety, transportability, or favorable policy selection. It is a within-project response to the poor neural Q approximation that checks the simulator's own fragility before adding architectural claims; code is src/sepsis_bootstrap.py.

Second sepsis series: serial sampling is not independent patients

A separate public whole-blood series, GSE54514, was fetched and parsed, with 163 measured GSM sample columns and 24,840 measured platform-probe expression rows. Sample metadata identify 96 sepsis-survivor blood collections, 31 nonsurvivor collections and 36 healthy collections, but many are serial daily samples. Across all 163 columns there are just 54 subject group IDs. Restricting to day one, the sepsis subset has 35 distinct patients: 26 labeled survivors and 9 nonsurvivors. This reduces repeated-measure leakage if a later descriptive model is trained, but the small nine-person outcome class limits any estimate. It is not a study of fluids or vasopressors and supplies no action sequence or state-matching information for ICU-Sepsis. We also read the first 48 finite numeric probe rows for these 35 day-one samples; for ILMN_1343291 the survivor-minus-nonsurvivor median expression is +0.340 in the provided matrix units. This is an arbitrary first-probe descriptive check, not a selected feature, significance test or mortality predictor. Its GSM accessions add another actual measured series for sepsis context only; do not count serial draws as independent patients, and do not claim a treatment-policy validation. Source: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE54514>; source gzip SHA256 and day-one GSM list in results/geo_independent.json, parse script src/geo_independent.py.

A perfect within-series mortality AUC that we do not accept as validation

A 35-patient day-one GSE54514 survival classification probe used the first 48 fully numeric matrix features and a five-fold stratified logistic evaluation. The computed out-of-fold AUC is 1.000 on nine nonsurvivors and 26 survivors. This number is not presented as a working mortality marker. The source orders all nine nonsurvivor day-one sample columns contiguously before all 26 survivor columns. Stratified folds split subjects but cannot split a batch or processing block if it tracks that label ordering. Moreover, the model is selected from an arbitrary convenience probe slice and has not been tested against a compatible outside platform; GSE65682's first probes use different platform IDs and share zero feature IDs with these chosen GSE54514 probes. A perfect AUC from only nine positive cases under these conditions is a signal to investigate batch, source annotation and normalization, not to announce a clinical result. The held-out accession IDs, source column indices, fold sizes and out-of-fold scores are all in results/sepsis_second_series.json, with code in src/sepsis_second_series.py. This sensitivity check adds no new accession count beyond the 35 already used from GSE54514 and no treatment-policy trajectories.

Availability audit for stronger ICU trajectories

The public MIMIC-III 100-patient demo was deliberately too small for a credible state-action sepsis evaluation. We checked a newer public MIMIC-Sepsis preprocessing framework, <https://github.com/yongh7/MIMIC-sepsis>, which describes sepsis trajectory tasks sourced from MIMIC-IV. Its README says using the patient database requires credentialed PhysioNet MIMIC-IV access, completion of research training, signing the data-use agreement and a locally loaded PostgreSQL copy. We cannot treat the framework's open code as an open patient-trajectory dataset. Its sepsis tasks also include mortality, stay length, vasopressor requirement and shock onset, which are not automatically off-policy treatment values. We use its methods as a future path only. In this repository the exact ICU-Sepsis MDP remains an aggregate simulator, the small public demo has real but incomplete repeated input events, and the two GEO series have blood labels but no treatment actions. The requested policy-data gate therefore remains zero. The source license for the 100-patient demo is the Open Data Commons Open Database License v1.0 at <https://physionet.org/content/mimiciii-demo/view-license/1.4/>; we distribute only derived aggregate counts pending careful application of the license and privacy scope to row-level exports.

Why a low regression error still gives weak control

We reproduced the one-hot Q-distillation network at the fixed original seed and audited its greedy action rankings against exact model Q values. Its terminal training loss was 0.000348, mean squared Q error on nonterminal state-action entries 0.000558, yet the chosen action exactly agreed with the optimal model action in only 5.75% of 713 nonterminal states. Many optimal action margins are small (median 0.00185), so a small Q error can swap rankings. Under exact model Q values, 61.9% of nonterminal states have a positive one-step Q shortfall under the neural action; the median shortfall is 0.00566, p95 0.03333. The model's initial-state distribution puts 95.8% mass on states where the chosen action differs from the exact best action. Weighted first-action regret under optimal continuation is 0.00486, far smaller than the full-policy value difference 0.08869, because later mistakes compound through the trajectory. This is not a clinical regret calculation or proof of a patient's treatment difference. It pinpoints a within-simulator failure mode that calls for rank-sensitive or occupancy-aware training instead of reporting a tiny regression loss as success. State-wise gaps and the ten largest initial-weighted contributors are in `results/sepsis_rank_audit.json`; the deterministic reconstruction is `src/sepsis_rank_audit.py`.

Rank-target pivot: a deliberately limited positive engineering check

A second within-project pivot targeted action ranking directly rather than regressing the precise Q value. We took each exact-model best action among 713 nonterminal states as a one-hot label and trained the same two-hidden-layer network at widths 32, 64 and 128 for 150 epochs. All three classify 100% of the 713 training states correctly and attain expected model return 0.875142, equal within numerical residual to exact value iteration and well above the earlier Q-regression network 0.786449. This fixes the implementation's action-ranking symptom but is not a treatment-policy discovery: every evaluated state was also a training state with exact-best labels, so this is supervised memorization of dynamic programming. Selecting an architecture after viewing these values is also not a valid held-out model comparison. No new patient treatment outcomes, intervention trajectories or new model regimes enter the test, and 0.875142 remains an in-model upper bound rather than a clinical survival estimate. The full width sweep and its exact labels, without choosing a supposed best model, are in `src/sepsis_rank_pivot.py` and `results/sepsis_rank_pivot.json`. Next meaningful validation would hold out an independently estimated MDP or acquire appropriately governed patient trajectories, not just improve same-table training accuracy.

The benchmark estimand is an estimated-model return, not observed survival improvement

The ICU-Sepsis authors explicitly present their MDP as a benchmark, not a medical practice guide: <https://arxiv.org/html/2406.05646>. They derive a 716-state, 25-action tabular system from approximately 17,000 MIMIC-III sepsis patients, discretizing treatment into combinations of intravenous-fluid and vasopressor doses and time into four-hour blocks. Three states represent termination, and the nonterminal state centroids encode 47 normalized clinical dimensions. The version we use is the released CSV transition model. We did not receive the individual training trajectories or counts attached to each state-action transition. The CSV transition matrix is a learned estimate, not a clinical randomization device and not a set of 716 actual people. The state index is a cluster assignment; changing an index in a neural input does not transport a regimen to a new patient with matching needs. The initial distribution and survival arrival reward define a mathematically clear but bounded in-model estimand.

Exact value iteration returns 0.875142 for that model, while the rounded published baseline is 0.88. The expert and uniformly random values in our same-table calculation round to 0.78. Agreement at two decimal places is a useful implementation check: transition indexing, reward placement and terminal handling plausibly match the source. It does not establish that a 0.875142 policy would have that survival frequency in a hospital. Our Q-regression neural controller valued at 0.786449 is inferior to the exact model solution by 0.088693 model-return units, but that number is not an estimated patient mortality increase. The rank-target controller closes the model-value gap only by consuming the model's exact best

action as every training label. It cannot be compared fairly to an RL agent that learns solely from sampled interactions, and calling it deep RL treatment improvement would overstate the evidence.

The action-support problem is especially important here. The source paper defines an admissible action in a state by an empirical transition-count threshold; sparse or unobserved actions are a central difficulty in constructing the MDP. Once a completed CSV transition row is provided, the row itself does not recover its source sample count, the uncertainty in its clustering, or patient-level confounding of treatment choice. Our Dirichlet experiment perturbs published nonzero probabilities under an arbitrary concentration. It therefore tests numerical sensitivity under a stated synthetic uncertainty class, rather than yielding uncertainty about clinical treatment effects. If one wanted a patient-level uncertainty interval, the resampling unit would have to be a patient trajectory, with the full original preprocessing repeated, including cohort eligibility, temporal binning, clustering, action threshold and transition estimation. We cannot perform that procedure from the aggregate model.

This distinction informs the next valid within-project test. A genuinely independent modeled comparison would fix algorithm, hyperparameters and selection rule before evaluating against an independently built transition system. If individual governed records were available, any off-policy estimate would need action propensities or other support diagnostics, outcome definition, held-out patients, and sensitivity analysis for measured and unmeasured confounding. The public 100-patient demo's 1,728 intervention-event rows over just 23 selected ICU stays cannot supply this in its present roughly screened form: multiple event rows come from a single stay, the screen is an administrative proxy, and the 25-action discretization plus four-hour matched state transitions has not been reconstructed. GEO blood expression likewise does not carry vasopressor/fluid actions. Without these checks, a model-return ranking remains exactly that, however precise the displayed decimals or long the simulation rollout.

Held-out state IDs expose the one-hot network's failure to transfer

The exact-label rank-target network above was fit on all 713 nonterminal state IDs. To check whether that success could survive even an internal holdout, we fixed three random state-ID splits before looking at results, each 570 training states and 143 test states, with network width 64 and 100 epochs. Labels for test states were not included in the training target. The network classified every training state correctly in all three splits, yet exact optimal-action agreement on the held-out states was only 0.126, 0.182 and 0.196. The within-split majority-action reference, computed from training labels and scored on the same test IDs, obtained 0.224, 0.196 and 0.154 respectively. Thus the held-out network is below that simple reference in two splits; small changes in the split also change this ordering. Mean exact-model Q shortfall per held-out state under its selected action was 0.00960, 0.00875 and 0.00955.

Evaluating each complete controller back in the unchanged MDP, including its correctly memorized training IDs, gives model return 0.856390, 0.854850 and 0.856915, below the exact 0.875142. Those full-policy values can mask the poor test-state classification, because many states are still trained and model occupancy is uneven. They are not independent simulation regimes or patient outcomes. The more direct explanation is representational: with a one-hot state input, an unseen ID activates an input coordinate whose associated first-layer weights have never received an example-specific gradient. The later layers share parameters but know nothing about that state's clinical similarity to a training state. This is a controlled diagnostic of missing shared state features and an honest negative after the all-state memorization result, not a general theorem that all neural ICU policies fail. The prespecified splits, state IDs, baseline predictions and calculation code are preserved in `results/sepsis_state_holdout.json` and `src/sepsis_state_holdout.py`. The next experiment should use meaningful state-cluster features or learn from independent transition trajectories, and must still distinguish model prediction from clinical effect.

References and audit links

- Choudhary K., Gupta D., Thomas P., ICU-Sepsis: A Benchmark MDP Built from Real Medical Data, RLC 2024. <https://arxiv.org/abs/2406.05646>.
- ICU-Sepsis model source and rounded baselines. <https://github.com/icu-sepsis/icu-sepsis>.